

SQL Fundamentals Exercise 2

Students Table

Question 1

--Question 1: List all distinct departments in the students table

```
SELECT DISTINCT department
FROM brightlearn.intelligence.students;
```

Add parameter

Table ▼ +

	^A _C department
1	IT
2	HR
3	Finance

Question 2

--Question 2: Get the average age of students per department:

```
SELECT department, AVG (age) AS student_age
FROM brightlearn.intelligence.students
GROUP BY department;
```

Add parameter

Table ▼ +

	^A _C department	1.2 student_age
1	IT	20.5
2	HR	22
3	Finance	23

Question 3

--Question 3: Show departments with more than 1 student:

```
SELECT department, COUNT(*) AS num_students
FROM brightlearn.intelligence.students
GROUP BY department
HAVING num_students > 1;
```

Add parameter

← Results 14 of 14 → Table ▾ +

	^A _C department	¹ ₂ ² ₃ num_students
1	IT	2
2	HR	2

Question 4

--Question 4: Get all students whose age is between 21 and 23:

```
SELECT student_id,  
       student_name,  
       age,  
       department  
FROM brightlearn.intelligence.students  
WHERE age BETWEEN 21 AND 23;
```

Add parameter

Table  

	¹ ₂ ³ student_id	^A _B ^C student_name	¹ ₂ ³ age	^A _B ^C department
1	2	Bob	22	HR
2	3	Charlie	21	IT
3	4	Diana	23	Finance
4	5	Eve	22	HR

Question 5

--Question 5: List all students in the IT or HR department who are older than 21:

```
SELECT student_id,  
       student_name,  
       age,  
       department  
FROM brightlearn.intelligence.students  
WHERE (department = 'IT' OR department = 'HR') AND age > 21;
```

Add parameter

Table  

	¹ ₂ ³ student_id	^A _B ^C student_name	¹ ₂ ³ age	^A _B ^C department
1	2	Bob	22	HR
2	5	Eve	22	HR

Courses table

Question 6

```
--Question 6: Show total credits per department, only for departments with more than 5 total credits
SELECT SUM(credits) AS total_credits
FROM brightlearn.intelligence.courses
GROUP BY department
HAVING total_credits > 5;
```

Add parameter

Table ▾ +

	1 ² ₃ total_credits
1	11

Question 7

```
--Question 7: List all courses that do not have 4 credits
SELECT course_id,
       course_name,
       department,
       credits
FROM brightlearn.intelligence.courses
WHERE NOT credits = 4;
```

Add parameter

Table ▾ +

	1 ² ₃ course_id	A ^B _C course_name	A ^B _C department	1 ² ₃ credits
1	101	SQL Basics	IT	3
2	104	Excel	Finance	2
3	105	Statistics	HR	3

Question 8

```
--Question 8: Show the top 3 courses by credits in descending order:
SELECT course_id,
       course_name,
       credits
FROM brightlearn.intelligence.courses
ORDER BY credits DESC
LIMIT 3;
```

Add parameter

Table  

	1 ² ₃ course_id	A ^B _C course_name	1 ² ₃ credits
1	102	Python	4
2	103	Data Science	4
3	101	SQL Basics	3

Enrollments table

Question 9

```
SELECT * FROM brightlearn.intelligence.enrollments;
```

```
--Question 9: Get the maximum, minimum and average grade across all enrollments
SELECT MAX (grade) AS max_grade,
       MIN (grade) AS min_grade,
       AVG (grade) AS avg_grade
FROM brightlearn.intelligence.enrollments;
```

Add parameter

Table  

	1 ² ₃ max_grade	1 ² ₃ min_grade	1.2 avg_grade
1	90	78	84.6

Question 10

```
--Count how many enrollments exist per course
SELECT course_id,
       COUNT (*) AS num_enrollments
FROM brightlearn.intelligence.enrollments
GROUP BY course_id;
```

Add parameter

Table  

	¹ ₃ course_id	¹ ₃ num_enrollments
1	101	1
2	102	1
3	103	1
4	104	1

Salaries table

Question 11

```
--Question 11: Find total salary and total bonus per department
SELECT department,
SUM (salary) AS total_salary,
SUM (bonus) AS total_bonus
FROM brightlearn.intelligence.salaries
GROUP BY department;
```

Add parameter

Table  

	 department	 total_salary	 total_bonus
1	IT	122000	10500
2	HR	109000	7500
3	Finance	70000	6000

Question 12

--Question 12: Show departments where average salary is above 55000

```
SELECT department,  
AVG (salary) AS avg_salary  
FROM brightlearn.intelligence.salaries  
GROUP BY department  
HAVING avg_salary > 55000;
```

Add parameter

Table  

	^A _C department	1.2 avg_salary
1	IT	61000
2	Finance	70000

Question 13

--Question 13: List employees whose salary plus bonus is greater than 60000

```
✓ SELECT employee_id,  
        employee_name,  
        salary,  
        bonus,  
        salary + bonus AS total_compensation  
FROM brightlearn.intelligence.salaries  
WHERE salary + bonus > 60000;
```

Add parameter

Table  

	¹ ₃ employee_id	^A _C employee_name	¹ ₃ salary	¹ ₃ bonus	¹ ₃ total_compensation
1	1	Tom	60000	5000	65000
2	3	Spike	70000	6000	76000
3	4	Tyke	62000	5500	67500

Projects table

Question 14

```
--Question 14: Show total and average budget per department. Only include departments with average budget above 70000
SELECT department,
SUM (budget) AS total_budget,
AVG (budget) AS avg_budget
FROM brightlearn.intelligence.projects
GROUP BY department
HAVING avg_budget > 70000;
```

Add parameter

Table  



	A _C ^B department	1 ₃ ² total_budget	1.2 avg_budget
1	IT	270000	135000
2	Finance	80000	80000

Question 15

```
--Question 15: List all projects with budgets between 50000 and 120000 excluding the Marketing department

SELECT project_id,
       project_name,
       department,
       budget
FROM brightlearn.intelligence.projects
WHERE budget BETWEEN 50000 AND 120000 AND department != 'Marketing';
```

Add parameter

Table  

	1 ₃ ² project_id	A _C ^B project_name	A _C ^B department	1 ₃ ² budget
1	1	AI App	IT	120000
2	2	Payroll System	Finance	80000
3	5	HR Portal	HR	50000